Multiply and Divide Using Scientific Notation

In the following exercises, multiply or divide and write your answer in decimal form.

408. $(2 \times 10^{5})(2 \times 10^{-9})$

409. $(3 \times 10^{2})(1 \times 10^{-5})$

410. $(1.6 \times 10^{-2})(5.2 \times 10^{-6})$

411. $(2.1 \times 10^{-4})(3.5 \times 10^{-2})$

412. $\frac{6 \times 10^{4}}{3 \times 10^{-2}}$

413. $\frac{8 \times 10^{6}}{4 \times 10^{-1}}$

414. $\frac{7 \times 10^{-2}}{1 \times 10^{-8}}$

415. $\frac{5 \times 10^{-3}}{1 \times 10^{-10}}$

Everyday Math

416. **Calories** In May 2010 the Food and Beverage Manufacturers pledged to reduce their products by 1.5 trillion calories by the end of 2015.

ⓐ Write 1.5 trillion in decimal notation.

ⓑ Write 1.5 trillion in scientific notation.

417. **Length of a year** The difference between the calendar year and the astronomical year is 0.000125 day.

ⓐ Write this number in scientific notation.

ⓑ How many years does it take for the difference to become 1 day?

418. **Calculator display** Many calculators automatically show answers in scientific notation if there are more digits than can fit in the calculator's display. To find the probability of getting a particular 5-card hand from a deck of cards, Mario divided 1 by $2{,}598{,}960$ and saw the answer 3.848×10^{-7}. Write the number in decimal notation.

419. **Calculator display** Many calculators automatically show answers in scientific notation if there are more digits than can fit in the calculator's display. To find the number of ways Barbara could make a collage with 6 of her 50 favorite photographs, she multiplied $50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45$. Her calculator gave the answer 1.1441304×10^{10}. Write the number in decimal notation.

Writing Exercises

420. ⓐ Explain the meaning of the exponent in the expression 2^{3}.

ⓑ Explain the meaning of the exponent in the expression 2^{-3}

421. When you convert a number from decimal notation to scientific notation, how do you know if the exponent will be positive or negative?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use the definition of a negative exponent.			
simplify expressions with integer exponents.			
convert from decimal notation to scientific notation.			
convert scientific notation to decimal form.			
multiply and divide using scientific notation.			

ⓑ *After looking at the checklist, do you think you are well prepared for the next section? Why or why not?*

10.6 Introduction to Factoring Polynomials

Learning Objectives

By the end of this section, you will be able to:

- Find the greatest common factor of two or more expressions
- Factor the greatest common factor from a polynomial

☑ **BE PREPARED** 10.15 Before you get started, take this readiness quiz.

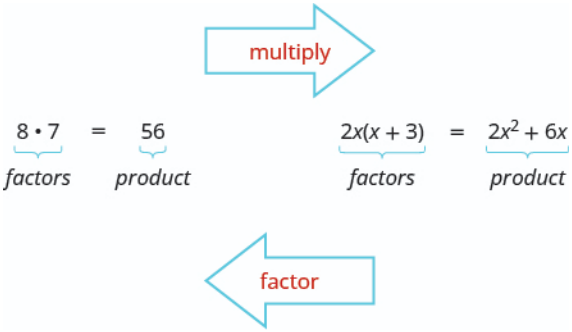
Factor 56 into primes.
If you missed this problem, review Example 2.48.

☑ **BE PREPARED** 10.16 Multiply: $-3(6a+11)$.
If you missed this problem, review Example 7.25.

☑ **BE PREPARED** 10.17 Multiply: $4x^2\left(x^2+3x-1\right)$.
If you missed this problem, review Example 10.32.

Find the Greatest Common Factor of Two or More Expressions

Earlier we multiplied factors together to get a product. Now, we will be reversing this process; we will start with a product and then break it down into its factors. Splitting a product into factors is called factoring.



In The Language of Algebra we factored numbers to find the least common multiple (LCM) of two or more numbers. Now we will factor expressions and find the *greatest common factor* of two or more expressions. The method we use is similar to what we used to find the LCM.

Greatest Common Factor

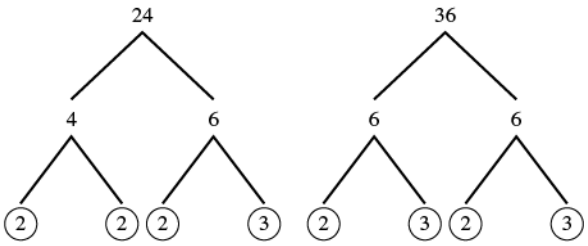
The greatest common factor (GCF) of two or more expressions is the largest expression that is a factor of all the expressions.

First we will find the greatest common factor of two numbers.

EXAMPLE 10.80

Find the greatest common factor of 24 and 36.

Solution

Step 1: Factor each coefficient into primes. Write all variables with exponents in expanded form.	Factor 24 and 36.	
Step 2: List all factors--matching common factors in a column.		$24 = 2 \cdot 2 \cdot 2 \cdot 3$ $36 = 2 \cdot 2 \cdot \quad 3 \cdot 3$

In each column, circle the common factors.	Circle the 2, 2, and 3 that are shared by both numbers.	$24 = 2 \cdot 2 \cdot 2 \cdot 3$ $36 = 2 \cdot 2 \cdot 3 \cdot 3$ $GCF = 2 \cdot 2 \cdot 3$ $GCF = 12$
Step 3: Bring down the common factors that all expressions share.	Bring down the 2, 2, 3 and then multiply.	
Step 4: Multiply the factors.		The GCF of 24 and 36 is 12.

Notice that since the GCF is a factor of both numbers, 24 and 36 can be written as multiples of 12.

$$24 = 12 \cdot 2$$
$$36 = 12 \cdot 3$$

TRY IT 10.159 Find the greatest common factor: $54, 36$.

TRY IT 10.160 Find the greatest common factor: $48, 80$.

In the previous example, we found the greatest common factor of constants. The greatest common factor of an algebraic expression can contain variables raised to powers along with coefficients. We summarize the steps we use to find the greatest common factor.

HOW TO

Find the greatest common factor.

Step 1. Factor each coefficient into primes. Write all variables with exponents in expanded form.
Step 2. List all factors—matching common factors in a column. In each column, circle the common factors.
Step 3. Bring down the common factors that all expressions share.
Step 4. Multiply the factors.

EXAMPLE 10.81

Find the greatest common factor of $5x$ and 15.

Solution

Factor each number into primes. Circle the common factors in each column. Bring down the common factors.	$5x = 5 \cdot x$ $15 = 3 \cdot 5$ $GCF = 5$
	The GCF of 5x and 15 is 5.

TRY IT 10.161 Find the greatest common factor: $7y, 14$.

TRY IT 10.162 Find the greatest common factor: $22,\ 11m$.

In the examples so far, the greatest common factor was a constant. In the next two examples we will get variables in the greatest common factor.

EXAMPLE 10.82

Find the greatest common factor of $12x^2$ and $18x^3$.

Solution

Factor each coefficient into primes and write the variables with exponents in expanded form. Circle the common factors in each column. Bring down the common factors. Multiply the factors.	$12x^2 = 2 \cdot 2 \cdot 3 \cdot x \cdot x$ $18x^3 = 2 \cdot 3 \cdot 3 \cdot x \cdot x \cdot x$ $\text{GCF} = 2 \cdot 3 \cdot x \cdot x$ $\text{GCF} = 6x^2$
	The GCF of $12x^2$ and $18x^3$ is $6x^2$

TRY IT 10.163 Find the greatest common factor: $16x^2,\ 24x^3$.

TRY IT 10.164 Find the greatest common factor: $27y^3,\ 18y^4$.

EXAMPLE 10.83

Find the greatest common factor of $14x^3,\ 8x^2,\ 10x$.

Solution

Factor each coefficient into primes and write the variables with exponents in expanded form. Circle the common factors in each column. Bring down the common factors. Multiply the factors.	$14x^3 = 2 \cdot 7 \cdot x \cdot x \cdot x$ $8x^2 = 2 \cdot 2 \cdot 2 \cdot x \cdot x$ $10x = 2 \cdot 5 \cdot x$ $\text{GCF} = 2 \cdot x$ $\text{GCF} = 2x$
	The GCF of $14x^3$ and $8x^2$, and$10x$ is $2x$

TRY IT 10.165 Find the greatest common factor: $21x^3,\ 9x^2,\ 15x$.

TRY IT 10.166 Find the greatest common factor: $25m^4,\ 35m^3,\ 20m^2$.

Factor the Greatest Common Factor from a Polynomial

Just like in arithmetic, where it is sometimes useful to represent a number in factored form (for example, 12 as $2 \cdot 6$ or $3 \cdot 4$), in algebra it can be useful to represent a polynomial in factored form. One way to do this is by finding the greatest common factor of all the terms. Remember that you can multiply a polynomial by a monomial as follows:

$$\begin{aligned} 2(x &+ 7) \text{ factors} \\ 2 \cdot x &+ 2 \cdot 7 \\ 2x &+ 14 \text{ product} \end{aligned}$$

Here, we will start with a product, like $2x + 14$, and end with its factors, $2(x + 7)$. To do this we apply the Distributive Property "in reverse".

Distributive Property

If $a,\ b,\ c$ are real numbers, then

$$a(b+c) = ab + ac \qquad \text{and} \qquad ab + ac = a(b+c)$$

The form on the left is used to multiply. The form on the right is used to factor.

So how do we use the Distributive Property to factor a polynomial? We find the GCF of all the terms and write the polynomial as a product!

EXAMPLE 10.84

Factor: $2x + 14$.

Solution

Step 1: Find the GCF of all the terms of the polynomial.	Find the GCF of 2x and 14.	$2x = 2 \cdot x$ $14 = 2 \cdot 7$ $\text{GCF} = 2$
Step 2: Rewrite each term as a product using the GCF.	Rewrite 2x and 14 as products of their GCF, 2. $2x = 2 \cdot x$ $14 = 2 \cdot 7$	$2x + 14$ $2 \cdot x + 2 \cdot 7$
Step 3: Use the Distributive Property 'in reverse' to factor the expression.		$2(x+7)$
Step 4: Check by multiplying the factors.		Check: $2(x+7)$ $2 \cdot x + 2 \cdot 7$ $2x + 14$ ✓

TRY IT 10.167 Factor: $4x + 12$.

TRY IT 10.168 Factor: $6a + 24$.

Notice that in Example 10.84, we used the word *factor* as both a noun and a verb:

Noun	7 is a factor of 14
Verb	factor 2 from $2x + 14$

HOW TO

Factor the greatest common factor from a polynomial.

Step 1. Find the GCF of all the terms of the polynomial.
Step 2. Rewrite each term as a product using the GCF.
Step 3. Use the Distributive Property 'in reverse' to factor the expression.
Step 4. Check by multiplying the factors.

EXAMPLE 10.85

Factor: $3a + 3$.

Solution

Find the GCF of $3a$, 3.

$$\begin{aligned} 3a &= 3 \cdot a \\ 3 &= 3 \\ \hline \text{GCF} &= 3 \end{aligned}$$

	$3a + 3$
Rewrite each term as a product using the GCF.	$3 \cdot a + 3 \cdot 1$
Use the Distributive Property 'in reverse' to factor the GCF.	$3(a + 1)$
Check by multiplying the factors to get the original polynomial.	
$3(a + 1)$ $3 \cdot a + 3 \cdot 1$ $3a + 3$ ✓	

TRY IT 10.169 Factor: $9a + 9$.

TRY IT 10.170 Factor: $11x + 11$.

The expressions in the next example have several factors in common. Remember to write the GCF as the product of all the common factors.

EXAMPLE 10.86

Factor: $12x - 60$.

Solution

Find the GCF of $12x$ and 60.

$$\begin{aligned} 12x &= 2 \cdot 2 \cdot 3 \cdot x \\ 60 &= 2 \cdot 2 \cdot 3 \cdot 5 \\ \hline \text{GCF} &= 2 \cdot 2 \cdot 3 \\ \text{GCF} &= 12 \end{aligned}$$

	$12x - 60$
Rewrite each term as a product using the GCF.	$12 \cdot x - 12 \cdot 5$
Factor the GCF.	$12(x - 5)$
Check by multiplying the factors.	
$12(x - 5)$ $12 \cdot x - 12 \cdot 5$ $12x - 60$ ✓	

TRY IT 10.171 Factor: $11x - 44$.

TRY IT 10.172 Factor: $13y - 52$.

Now we'll factor the greatest common factor from a trinomial. We start by finding the GCF of all three terms.

EXAMPLE 10.87

Factor: $3y^2 + 6y + 9$.

Solution

Find the GCF of $3y^2$, $6y$, and 9.

$$\begin{aligned} 3y^2 &= 3 \cdot y \cdot y \\ 6y &= 2 \cdot 3 \cdot y \\ 9 &= 3 \cdot 3 \\ \hline \text{GCF} &= 3 \end{aligned}$$

	$3y^2 + 6y + 9$
Rewrite each term as a product using the GCF.	$3 \cdot y^2 + 3 \cdot 2y + 3 \cdot 3$
Factor the GCF.	$3(y^2 + 2y + 3)$
Check by multiplying.	
$3(y^2 + 2y + 3)$ $3 \cdot y^2 + 3 \cdot 2y + 3 \cdot 3$ $3y^2 + 6y + 9$ ✓	

TRY IT 10.173 Factor: $4y^2 + 8y + 12$.

TRY IT 10.174 Factor: $6x^2 + 42x - 12$.

In the next example, we factor a variable from a binomial.

EXAMPLE 10.88

Factor: $6x^2 + 5x$.

Solution

	$6x^2 + 5x$
Find the GCF of $6x^2$ and $5x$ and the math that goes with it.	$6x^2 = 2 \cdot 3 \cdot x \cdot x$ $5x = 5 \cdot x$ GCF $= x$
Rewrite each term as a product.	$x \cdot 6x + x \cdot 5$
Factor the GCF.	$x(6x + 5)$

Check by multiplying.

$x(6x + 5)$
$x \cdot 6x + x \cdot 5$
$6x^2 + 5x$✓

TRY IT 10.175 Factor: $9x^2 + 7x$.

TRY IT 10.176 Factor: $5a^2 - 12a$.

When there are several common factors, as we'll see in the next two examples, good organization and neat work helps!

EXAMPLE 10.89

Factor: $4x^3 - 20x^2$.

Solution

Find the GCF of $4x^3$ and $20x^2$.

$4x^3 = 2 \cdot 2 \cdot x \cdot x \cdot x$
$20x^2 = 2 \cdot 2 \cdot 5 \cdot x \cdot x$
$\text{GCF} = 2 \cdot 2 \cdot x \cdot x$
$\text{GCF} = 4x^2$

		$4x^3 - 20x^2$
Rewrite each term.		$4x^2 \cdot x - 4x^2 \cdot 5$
Factor the GCF.		$4x^2(x - 5)$
Check.	$4x^2(x - 5)$ $4x^2 \cdot x - 4x^2 \cdot 5$ $4x^3 - 20x^2$ ✓	

TRY IT 10.177 Factor: $2x^3 + 12x^2$.

TRY IT 10.178 Factor: $6y^3 - 15y^2$.

EXAMPLE 10.90

Factor: $21y^2 + 35y$.

Solution

Find the GCF of $21y^2$ and $35y$	$21y^2 = 3 \cdot 7 \cdot y \cdot y$ $35y = 5 \cdot 7 \cdot y$ $GCF = 7 \cdot y$ $GCF = 7y$	
		$21y^2 + 35y$
Rewrite each term.		$7y \cdot 3y + 7y \cdot 5$
Factor the GCF.		$7y(3y + 5)$

TRY IT 10.179 Factor: $18y^2 + 63y$.

TRY IT 10.180 Factor: $32k^2 + 56k$.

EXAMPLE 10.91

Factor: $14x^3 + 8x^2 - 10x$.

Solution

Previously, we found the GCF of $14x^3$, $8x^2$, and $10x$ to be $2x$.

	$14x^3 + 8x^2 - 10x$
Rewrite each term using the GCF, 2x.	$2x \cdot 7x^2 + 2x \cdot 4x - 2x \cdot 5$
Factor the GCF.	$2x(7x^2 + 4x - 5)$
Check. $2x\,(7x^2 + 4x - 5)$ $2x \cdot 7x^2 + 2x \cdot 4x - 2x \cdot 5$ $14x^3 + 8x^2 - 10x$ ✓	

TRY IT 10.181 Factor: $18y^3 - 6y^2 - 24y$.

TRY IT 10.182 Factor: $16x^3 + 8x^2 - 12x$.

When the leading coefficient, the coefficient of the first term, is negative, we factor the negative out as part of the GCF.

EXAMPLE 10.92

Factor: $-9y - 27$.

Solution

When the leading coefficient is negative, the GCF will be negative. Ignoring the signs of the terms, we first find the GCF of $9y$ and 27 is 9.	$9y = 3 \cdot 3 \cdot y$ $27 = 3 \cdot 3 \cdot 3$ $\text{GCF} = 3 \cdot 3$ $\text{GCF} = 9$
Since the expression $-9y-27$ has a negative leading coefficient, we use -9 as the GCF.	
	$-9y - 27$
Rewrite each term using the GCF.	$-9 \cdot y + (-9) \cdot 3$
Factor the GCF.	$-9(y + 3)$
Check. $-9(y + 3)$ $-9 \cdot y + (-9) \cdot 3$ $-9y - 27$ ✓	

TRY IT 10.183 Factor: $-5y - 35$.

TRY IT 10.184 Factor: $-16z - 56$.

Pay close attention to the signs of the terms in the next example.

EXAMPLE 10.93

Factor: $-4a^2 + 16a$.

Solution

The leading coefficient is negative, so the GCF will be negative.	
	$4a^2 = 2 \cdot 2 \cdot a \cdot a$ $16a = 2 \cdot 2 \cdot 2 \cdot 2 \cdot a$ $\text{GCF} = 2 \cdot 2 \cdot a$ $\text{GCF} = 4a$
Since the leading coefficient is negative, the GCF is negative, $-4a$.	
	$-4a^2 + 16a$
Rewrite each term.	$-4a \cdot a - (-4a) \cdot 4$
Factor the GCF.	$-4a(a - 4)$
Check on your own by multiplying.	

TRY IT 10.185 Factor: $-7a^2 + 21a$.

TRY IT 10.186 Factor: $-6x^2 + x$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Factor GCF (https://www.openstax.org/l/24factorgcf)
Factor a Binomial (https://www.openstax.org/l/24factorbinomi)
Identify GCF (https://www.openstax.org/l/24identifygcf)

SECTION 10.6 EXERCISES

Practice Makes Perfect

Find the Greatest Common Factor of Two or More Expressions

In the following exercises, find the greatest common factor.

422. $40, 56$

423. $45, 75$

424. $72, 162$

425. $150, 275$

426. $3x, 12$

427. $4y, 28$

428. $10a, 50$

429. $5b, 30$

430. $16y,\ 24y^2$

431. $9x,\ 15x^2$

432. $18m^3,\ 36m^2$

433. $12p^4,\ 48p^3$

434. $10x,\ 25x^2,\ 15x^3$

435. $18a,\ 6a^2,\ 22a^3$

436. $24u,\ 6u^2,\ 30u^3$

437. $40y,\ 10y^2,\ 90y^3$

438. $15a^4,\ 9a^5,\ 21a^6$

439. $35x^3,\ 10x^4,\ 5x^5$

440. $27y^2,\ 45y^3,\ 9y^4$

441. $14b^2,\ 35b^3,\ 63b^4$

Factor the Greatest Common Factor from a Polynomial

In the following exercises, factor the greatest common factor from each polynomial.

442. $2x + 8$

443. $5y + 15$

444. $3a - 24$

445. $4b - 20$

446. $9y - 9$

447. $7x - 7$

448. $5m^2 + 20m + 35$

449. $3n^2 + 21n + 12$

450. $8p^2 + 32p + 48$

451. $6q^2 + 30q + 42$

452. $8q^2 + 15q$

453. $9c^2 + 22c$

454. $13k^2 + 5k$

455. $17x^2 + 7x$

456. $5c^2 + 9c$

457. $4q^2 + 7q$

458. $5p^2 + 25p$

459. $3r^2 + 27r$

460. $24q^2 - 12q$

461. $30u^2 - 10u$

462. $yz + 4z$

463. $ab + 8b$

464. $60x - 6x^3$

465. $55y - 11y^4$

466. $48r^4 - 12r^3$

467. $45c^3 - 15c^2$

468. $4a^3 - 4ab^2$

469. $6c^3 - 6cd^2$

470. $30u^3 + 80u^2$

471. $48x^3 + 72x^2$

472. $120y^6 + 48y^4$

473. $144a^6 + 90a^3$

474. $4q^2 + 24q + 28$

475. $10y^2 + 50y + 40$

476. $15z^2 - 30z - 90$

477. $12u^2 - 36u - 108$

478. $3a^4 - 24a^3 + 18a^2$

479. $5p^4 - 20p^3 - 15p^2$

480. $11x^6 + 44x^5 - 121x^4$

481. $8c^5 + 40c^4 - 56c^3$

482. $-3n - 24$

483. $-7p - 84$

484. $-15a^2 - 40a$

485. $-18b^2 - 66b$

486. $-10y^3 + 60y^2$

487. $-8a^3 + 32a^2$ **488**. $-4u^5 + 56u^3$ **489**. $-9b^5 + 63b^3$

Everyday Math

490. **Revenue** A manufacturer of microwave ovens has found that the revenue received from selling microwaves a cost of p dollars each is given by the polynomial $-5p^2 + 150p$. Factor the greatest common factor from this polynomial.

491. **Height of a baseball** The height of a baseball hit with velocity 80 feet/second at 4 feet above ground level is $-16t^2 + 80t + 4$, with $t =$ the number of seconds since it was hit. Factor the greatest common factor from this polynomial.

Writing Exercises

492. The greatest common factor of 36 and 60 is 12. Explain what this means.

493. What is the GCF of y^4, y^5, and y^{10}? Write a general rule that tells how to find the GCF of y^a, y^b, and y^c.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
find the greatest common factor of two or more expressions.			
factor the greatest common factor from a polynomial.			

ⓑ *Overall, after looking at the checklist, do you think you are well-prepared for the next Chapter? Why or why not?*